MM-Predict: Risk predictions for progression from MGUS to multiple myeloma enhanced by polygenic risk scores

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***Abstract*—Multiple myeloma (MM) is a hematologic malignancy that develops from the premalignant condition monoclonal gammopathy of undetermined significance (MGUS). Accurate prediction of progression from MGUS to MM is critical for enabling early, targeted intervention. While established clinical models, like the PANGEA model [1] rely on biomarkers such as M-protein concentration and serum free light chain ratio, they do not account for an individual's inherited genetic risk. In this paper, we investigate whether polygenic risk scores (PRS) can improve MM progression prediction when combined with traditional clinical features. Using a synthetic cohort of 2,000 individuals derived from the FinnGen dataset, we train and compare three survival models each evaluated with and without polygenic risk information. We find a consistent improvement in concordance index when PRS is included, rising from 0.690 to 0.709 in our best-performing model, with PRS ranking as the second strongest predictor after M-protein concentration. Although the use of synthetic data limits generalizability, our work provides a methodological framework for PRS-augmented survival modelling that future researchers can apply to real-world cohorts to further validate these findings.**

***Index Terms*—Smoldering multiple myeloma, Systematic model comparison, Polygenic risk score, Cox PH Model, Risk stratification, FinnGen dataset**

I. Introduction

Multiple myeloma is a type of blood cancer that develops in plasma cells in the bone marrow [2]. With a worldwide incidence rate of 2.1 per 100,000 new cases every year, its early diagnosis and effective treatment are of high interest [3]. Usually, Multiple myeloma evolves from a premalignant condition called monoclonal gammopathy of undetermined significance, MGUS. This is often followed by an intermediate condition called smoldering multiple myeloma that elevates progression risks to MM further [2]. Given the serious side effects and costs associated with treatment and the benefits of treatment in early stages of the illness the importance of individualized care is on the rise. Traditional models that predict an individual's risk of progressing from MGUS or smoldering MM to MM heavily rely on traditional biomarkers, such as the M-protein concentration in the patient's blood [4]. However, these models do not account for an individual's inherited genetic risk.

Polygenic risk scores can capture this inherited predisposition of an individual. These scores are calculated by aggregating the effect of many genetic variants across the genome into a single measure and have been shown to improve prediction accuracies in many areas of medical risk modelling [5]. Additionally, studies have shown that multiple myeloma risk has a strong genetic background and that polygenic risk scores can contribute to a MM risk prediction [3]. We investigate the contribution of polygenic risk factors in MM progression risk prediction models. More specifically, this paper compares a selection of statistical and machine learning models' performance when using only traditional biomarkers and when additionally having access to an individual's polygenic risk score.

This work builds on the PANGEA study, which developed a clinical risk prediction model for MGUS progression to multiple myeloma using traditional biomarkers [1]. PANGEA established a strong baseline for clinical risk stratification, but did not incorporate genetic risk factors. We extend this line of investigation by introducing polygenic risk scores as an additional covariate, using PANGEA's clinical feature set as the foundation for better comparability. Just like the PANGEA team we made our model publicly available at https://mm-predict.jan-berndt.de/ using a simple web interface to make our findings easily accessible for research purposes.

Since we work with synthetic data with limited generalizability, this work is intended as a methodological foundation for systematic comparison, designed so that future researchers can substitute real-world cohorts and draw context-specific conclusions. Nonetheless, our results consistently show improved model discrimination when polygenic risk scores are incorporated, providing early evidence for their prognostic value in myeloma progression risk modelling.

II. Clinical Background

Multiple myeloma (MM) is a malignant hematologic cancer characterized by the clonal growth of plasma cells in the bone marrow. It remains largely incurable despite advances in treatment and new therapeutic options being approved every year [6]. MM typically evolves through a well-defined precursor condition beginning with Monoclonal Gammopathy of Undetermined Significance (MGUS) and, in some cases, progresses through an intermediate stage known as smoldering multiple myeloma (SMM).

A. MGUS

MGUS is an asymptomatic premalignant condition defined by the presence of a serum monoclonal protein (M-protein) below 3 g/dL, fewer than 10% clonal plasma cells in the bone marrow, and the absence of end-organ damage. It is quite common in the general population, with approximately 3% of adults over 50 having the condition. The prevalence rises even more in population groups of higher age. While MGUS itself requires no treatment, it carries a risk of progression to MM or a related malignancy of approximately 1% per year [7].

B. SMM

SMM represents an intermediate stage between MGUS and active MM. It can be defined by higher tumor burden and either a serum M-protein of $\geq 3\frac{g}{d}L$ or $\geq 10\%$ clonal plasma cells in the bone marrow. It still lacks the organ damage that characterizes active multiple myeloma. The risk of progression from SMM to MM is significantly higher than from MGUS, estimated at around 10% per year in the first five years following diagnosis [8]. Because of this high progression risk patients with smoldering MM are of high interest for early interventions. The higher prevalence of MGUS and the historic lack of standardized diagnostic criteria for SMM lead to MGUS often being the clinically established starting point for progression modelling in the field [9].

C. Diagnosis

In 2014 the way multiple myeloma is diagnosed was updated. Previously, active MM was diagnosed when clonal plasma cell proliferation leads to end-organ damage. This definition has been amended by the International Myeloma Working Group (IMWG) to include additional biomarkers. More concretely, MM is diagnosed once clonal bone marrow plasma cells occupy $\geq 60\%$ of the bone marrow, the serum free light chain ratio is bigger 100, or more than one focal lesion on MRI. This push for more personalized diagnosis based on biomarkers before serious organ damage occurs is a strong sign for the potential of individualized diagnosis and treatment. Despite effective therapies including proteasome inhibitors, immunomodulatory drugs, and CAR-T cell therapy, MM remains incurable in most patients, with a median overall survival that varies widely by disease stage and patient profile [2].

III. Methods

A. Data and Cohort

FinnGen is a dataset containing 224,737 whole genome sequences of Finnish individuals combined with phenotypes extracted from electronic health records [10]. Out of data protection concerns, we did not use the actual dataset to conduct the study. Instead, we opted for a set of 2000 synthetically generated individuals based on the FinnGen dataset. The generated participants were based on real individuals diagnosed with MGUS. From the FinnGen data, it was not always clearly distinguishable whether individuals with an MGUS diagnosis already qualified as having SMM. We assumed MGUS unless the data explicitly stated otherwise.

Each participant is labeled with a status variable indicating whether the patient was diagnosed with multiple myeloma before the FinnGen's cutoff date. A time column contains the duration either between the patient's MGUS and MM diagnoses, or from the patients MGUS diagnosis up to right-censoring. Note that in our case, a patient dying without having developed MM also qualifies as a right-censoring event.

Before generating the synthetic dataset, PRS scores were calculated for the actual FinnGen participants using PRS-CS [11] based on the genome-wide association study conducted by Went et al. in 2024 [12]. Synthetic data points were then generated with an associated polygenic score. The final dataset given to us contained 2000 synthetically generated participants and the columns as described in Table I.

TABLE I
COLUMNS OF AVAILABLE DATA AND THEIR MEANING

Ancestry	European / African American
Polygenic Score	Normalized Polygenic Score relative to whole cohort
Age	Age in years
M-Spike	M-Spike protein blood concentration in g/dL
SFLC Ratio	Serum free light chain ratio of involved / uninvolved chains
Creatinine	Creatinine blood concentration in mg/dL
LP	Liability prediction combining both clinical and genetic risk
BMPC	Bone marrow plasma cell percentage
HGB	Hemoglobin levels
PGS Bin	Low / medium / high based on polygenic score
Clinical Risk	General clinical risk factor based on demographics
Time	Time in months until either MM diagnosis or right-censoring
Status	Whether participant was diagnosed with MM or right-censored

For handling missing data and imputation, we first inspected data completeness per feature and per sample. You can find the results in Table II. We found high availability across all features and samples. The only feature requiring special consideration is the *bone marrow plasma cell percentage*. This feature describes the result of an invasive bone marrow examination that usually is only performed by medical experts when a high MM risk is suspected. One key finding of the similar PANGEA study [1] was that bone marrow biopsies are not required for accurate mm progression risk prediction. For better comparability to this paper and the PANGEA model we do not keep this covariate for our models. Instead, we included an indicator covariate *was_bmpc_measured* that equals 1 if a bone marrow measurement was performed and 0 otherwise. This way, the influence of bone marrow measurements can be judged more clearly during feature selection. The only other feature with missing data is the blood's hemoglobin concentration *hgb*. For this, we use a simple but well established imputation approach: KNN imputation [13].

TABLE II
ANALYSIS OF MISSING VALUES PER FEATURE AND SAMPLE

Missing rate	Features		Samples	
	n	%	n	%
[0.0, 0.2]	10	90.9	2000	100.0
(0.2, 0.4]	1	9.1	0	0.0
(0.4, 0.6]	0	0.0	0	0.0
(0.6, 0.8]	0	0.0	0	0.0
(0.8, 1.0]	0	0.0	0	0.0

B. Feature Selection and engineering

Many risk modelling approaches in the field of personalized medicine deal with high-dimensional, often sparse and complex data. Given this and the often times small cohort size, careful feature engineering and selection are essential to avoid overfitting and to ensure the model's interpretability. This mismatch of dimensionality and number of sample is often referenced as *'curse of dimensionality'* in the field [14]. Compared to similar risk modelling applications in personalized medicine our data is of low dimensionality and has few missing values. Even so, understanding the information gain attributed to each feature is important. This way the model's inner workings and limitation can be understood better.

For comparing the feature's contributions to the predictions we used the commonly employed penalized Cox regression method. This regression method takes the partial likelihood for all n subjects

$$l(\beta) = \prod_{i=0}^{n} l_i(\beta) \tag{1}$$

where $l_{i(\beta)}$ represents the *i'th* individual's partial likelihood contribution. For each parameter β_i we then add a penalty $P_\gamma(\beta_i)$ to this partial likelihood function [15]. This forces the regression to limit the total size of its parameters β forming a new objective function

$$Q(\beta) = l(\beta) - \sum_{j=1}^{p} P_\gamma(\beta_j) \tag{2}$$

When this function is maximized during Cox ph model fitting using regression the contribution of uninformative features shrunk [16]. In the package we use, *lifelines* [17], the penalty function is chosen as a weighted sum of L_1 and L_2 regularization like this:

$$P_\gamma(\beta_j) = \alpha \cdot \sum_j |\beta_j| + \frac{1-\alpha}{2} \cdot \sum_j \beta_j^2 \quad (3)$$

For identifying the most important features in our dataset we performed hyperparameter selection via grid search over the regularization strength $\lambda \in \{0.001, 0.01, 0.1\}$ and the elastic net mixing parameter $\alpha \in \{0.0, 0.1, ..., 1.0\}$. For each parameter combination, model performance was estimated using 10-fold cross-validation, with the concordance index (C-index) as the evaluation metric. The combination yielding the highest mean C-index across folds was selected for the final model. This model then was used for interpreting each parameter's information contribution. The code used for performing this systematic comparison can be found publicly available at https://github.com/jafber/mm-predict.
After identifying the features of interest we performed normalization on all continuous features to ensure that the Cox model's parameters do not cause numerical problems. We endeavored to make our final model publicly available similar to the way the PANGEA research team made their models available for public use on a simple web interface [1]. To make our model easier to use in such a context, we created a new feature *pgs_bin*. This feature groups patients into a low, medium and high risk group based on PRS percentile. Before doing this we confirmed that the information loss caused by this simplification was acceptable.

C. Model development

In recent years, machine and deep learning techniques are being increasingly used in the field of personalized medicine. Their ability to find complex interdependencies between features during training can improve predictive accuracy in comparison to traditional models. Even so, this improvement comes at the cost of reduced interpretability, which is very important in many high-stakes medical environments. In this paper we seek to compare the predictive power of more sophisticated machine learning models with traditional, proven models. More specifically, we chose the Cox proportional hazard model as a baseline model since it is well established, widely used and easily interpretable [18].
Additionally, we hypothesize that polygenic risk scores can have complex dependencies to other features that may not be captured by the Cox proportional hazard model. Therefore, we also train machine learning models that can reflect these potentially interconnected effects between features to test this hypothesis. For this we chose gradient boosted machines and random survival forests because of their successful application in similar projects [16], [18].

a) *Cox proportional hazard model:* The Cox proportional hazards model is one of the most common regression approaches in survival analysis [15] and serves as a foundation for many advanced deep learning models [18]. A key assumption is the proportionality of hazards, meaning that the hazard ratio between any two patients remains constant over time. If the cohort consists of fundamentally different subgroups whose hazards evolve differently over time, this assumption may be violated. In practice, this can be addressed through stratification and by including all major prognostic factors as covariates.
Following [15] the Cox model defines a hazard function for subject i as

$$h_{i(t)} = h_0(t) \exp(x_i^T \beta) \quad (4)$$

where β is the coefficient vector for the model, h_0 is the baseline hazard and $\exp(x_i^T \beta)$ is the risk for a subject with covariates x_i. The coefficients β can be obtained by maximizing the partial likelihood or even the penalized partial likelihood as described in Section III.B. Note that the independence of covariates that is explicitly assumed in the definition of the hazard function.

b) *Random survival forest:* Random survival forests (RSF) [19] extend the random forest framework to survival data. As with standard random forests, an ensemble of decision trees is built on bootstrap samples of the training data, with each split considering a random subset of covariates. The main difference for using this with survival data is the split criterion and the leaf node estimator: rather than minimizing variance or impurity, splits are chosen to maximize the survival difference between child nodes, typically measured by the log-rank statistic. RSFs make no parametric assumptions. In particular, they do not require proportional hazards, making them well-suited for testing for complex feature correlations. The hyperparameter $n_{\text{estimators}}$ defines how many trees are used to construct the random survival forest. To systematically test this model we used 10-fold cross validation tuning $n_{\text{estimators}}$ both with and without the polygenic risk scores in the features.

c) *Gradient Boosted machine:* Gradient boosted machines (GBM) [20] are another tree-based ensemble method, but in contrast to random forests, trees are built sequentially rather than in parallel. Each tree is fit to the residuals of the previous ensemble, gradually correcting errors and improving the overall prediction performance. The key hyperparameters are the number of trees ($n_{\text{estimators}}$), the learning rate η, which scales the contribution of each tree, and the maximum tree depth, which controls model complexity. The learning rate and number of trees trade off against each other. A smaller learning rate requires more trees but often has better generalization. As with RSFs, GBMs make no proportional hazards assumption and naturally capture non-linear effects and feature interactions. We tested a selection of hyperparameter combinations of $n_{\text{estimators}}$, learning rate and max_depth using 10-fold cv to find the best model for our data once with and once without polygenic risk scores.

D. Evaluation and Validation

For comparing models we use the concordance index. This metric measures the probability that, for a randomly selected pair of subjects, the model correctly assigns a higher risk score to the subject who experiences the event first. The C-index ranges from 0.5, indicating no predictive ability, to 1.0, indicating perfect prediction for a set up subjects. Since this score only tests how well a model can sort subject's risk relative to one another, it is not suited absolute predictions and comparisons. We however only care about the relative performance of one model compared to another, so this metric serves its purpose for us.

IV. Results

A. Feature selection and engineering

The grid search we performed for feature selection over the regularization strength λ and the elastic net mixing parameter α for the best Cox ph model showed optimal model performance for $\lambda = 0.01$ and $\alpha = 0.1$. Fig. 1 shows the hazard ratios in the final Cox ph model for each of the features and their respective 95% confidence intervals. Notably, the hazard ratio of *hgb* is below one, the hazard ratios of *ancestry* and of *was_bmpc_measured* very close to one and all the other ratios greater than one.

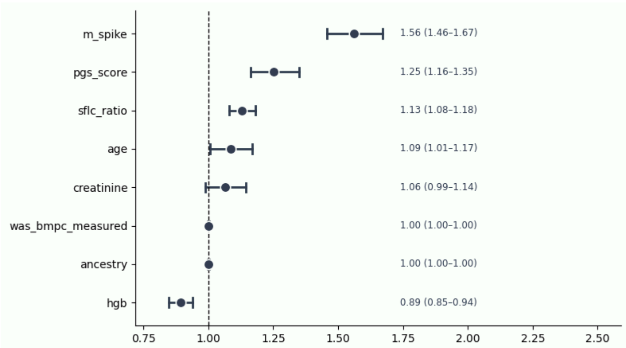

Fig. 1. Hazard ratio per 1 standard deviation by feature, with 95% CIs

B. Model development

As described in Section III.C we performed exhaustive grid hyperparameter search for random survival forests, gradient boosted machines and the standard Cox proportional hazard model. Each of the models was optimized both for the dataset with and without polygenic risk information. The respective concordance indices can be found in Fig. 2.

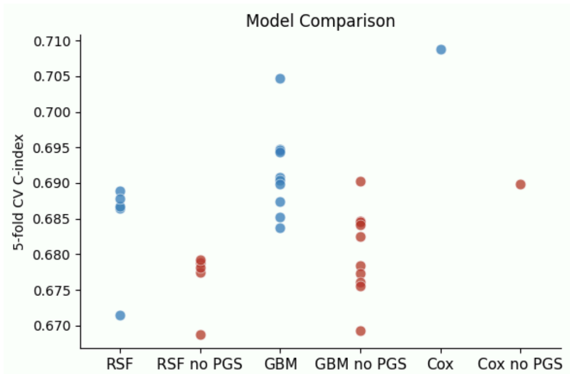

Fig. 2. Comparison of Random survival forest, Gradient boosted machine and Cox PH models with and without polygenic risk scores respectively

For the dataset including polygenic risk scores, the best 5-fold c-index achieved per model type was 0.689 for the RSF, 0.704 for the GBM and 0.709 for the Cox model. Without polygenic risk score, the best indices were 0.679 for the RSF, 0.690 for the GBM and 0.690 for the Cox model.

C. MM-Predict calculator

To share our findings, we implemented a web application to make MGUS to MM progression predictions easily accessible to both patients and doctors. Just as in the PANGEA calculator [21], the app provides a form to put in SFLC ratio, M-Spike, Creatinine and Age biomarkers. Optionally, users can provide a genetic risk factor to make predictions more accurate. To see the full application, visit mm-predict.jan-berndt.de. The code is publicly available under MIT licence at https://github.com/jafber/mm-predict. Please note, that the model should be used for research purposes only and does not have the underlying data sets nor the scientific rigor needed for actual clinical application yet.

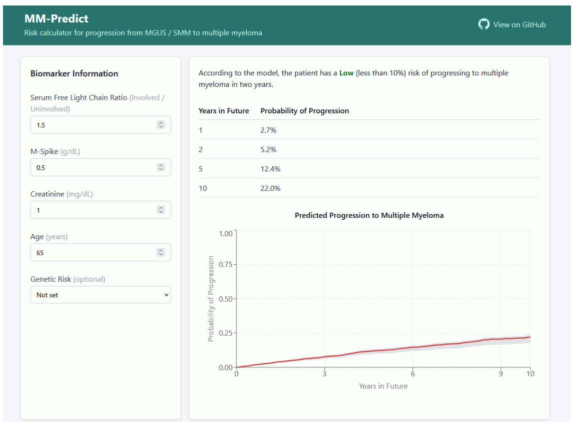

Fig. 3. MM-Predict calculator interface found at mm-predict.jan-berndt.de

V. Discussion

A. Feature Selection and Engineering

The five features *m_spike*, *pgs_score*, *slfc_ratio*, *age* and *creatinine* exhibit a hazard ratio that is greater or equal to one on average (see Fig. 1). This matches our expectations of those biomarkers: higher values increase the MM progression risk. Especially the high importance of the M-spike protein agrees with the consensus on multiple myeloma diagnostics criteria [22], [23]. The hazard ratio of 0.89 for hemoglobin *hgb* is consistent with established findings that lower hemoglobin levels reflect end-organ damage, which is associated with increased risk of progression to multiple myeloma [23].

a) *Effect of ancestry feature:* It is well established that multiple myeloma is more prevalent in black people compared to white people. A 2010 study found that in the US, MM incidence is twice as high in blacks [24]. Previous work suggests that this disparity can be explained by higher MGUS prevalence among black populations, whereas the progression from MGUS to MM itself is not affected by ethnicity [25].

This evidence is corroborated by our findings. The p value for ancestry in the Cox survival model was 0.99, indicating that ancestry does not significantly impact MM progression risk. Because of this lack of significance, we chose not to include ancestry as a feature in our final model.

b) *Effect of bone marrow plasma cell percentage:* According to our results the feature if the bone marrow plasma cell percentage contributes nearly no information to the risk prediction. Traditionally, this biomarker can have great diagnostic value [26]. Even so, it has been hypothesized that in presence of other modern predictive factors, like genetical analysis of patients inherent risk, this biomarker might lose its importance in risk prediction [26]. We suspect, that in our case the little predictive value stems from the way our synthetic data was modeled and generated. Further research on other cohorts would be needed to inspect the influence of bone *marrow plasma cell percentages* in predictive models.

c) *Significance of PRS Scores in Progression Prediction:*

Previous work showed that genetic factors have a strong influence on multiple myeloma risk. Canzian et al. were able to develop a PRS where study participants in the highest score quintile had a risk of developing MM 3.44 times higher than for participants in the lowest quintile [3].

Since our dataset only consisted of MGUS patients without a healthy control group, a direct comparison to the Canzian et al. study is not possible.

However, our results clearly corroborate that polygenic scores are a significant factor for predicting MM. In the MM Predict (PRS) model, the polygenic score had a hazard ratio of 1.25, meaning that per standard deviation in our polygenic score MM progression hazard increases by a factor of 1.25. This made PRS the second strongest MM risk indicator in our feature set, outweighed only by M-Spike protein concentration. We were able to improve our base model's C-Statistic from an average of 0.690 to 0.709 by adding in PGS.

B. Model development

The Cox model showed the highest c index both when predicting on data with and without polygenic risk information. Out of the three model types we chose to move forward with the Cox model, since it was both the most accurate and most performant.
However, both the random forest and gradient boosting estimators performed close to the Cox models when the right hyperparameters were chosen, with C-index differences in the range of ~1% between the highest performing models per type. This suggests that the type of model does not significantly influence prediction accuracy on our dataset. We hypothesize that this stems from the artificial generation of our dataset that does not necessarily model complex inter-feature dependencies which machine learning models excel at.

C. Comparison to PANGEA

To enable a direct comparison, we designed our non-PRS model to operate on the same input features as the PANGEA model [21], predicting progression from MGUS to MM using age, M-protein, serum free light chain ratio, and creatinine from a single timepoint. As shown in Fig. 4, both of our models achieved lower concordance indices than PANGEA, though our 95% bootstrap confidence intervals were considerably narrower than PANGEA's wide range of 0.586–0.938, suggesting greater stability in our estimates. Several factors likely explain PANGEA's superior discriminative performance. First, the PANGEA study incorporated up to three longitudinal examinations per patient, enabling the model to capture temporal trends in biomarker trajectories rather than relying on a single snapshot. Second, PANGEA was trained on a larger real-world cohort, whereas our dataset was smaller and synthetic. Synthetic data generation may not fully reproduce the statistical dependencies present in real patient populations, potentially limiting our model's ability to learn clinically meaningful patterns.

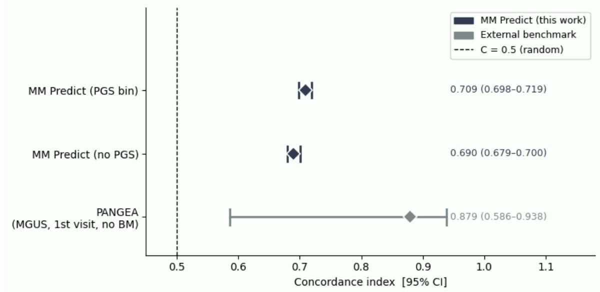

Fig. 4. Comparison of our models and the PANGEA model [1]

VI. Conclusion and Outlook

In this work, we investigated whether polygenic risk scores can improve the prediction of progression from MGUS to multiple myeloma when combined with established clinical biomarkers. Using a synthetic cohort derived from the FinnGen dataset, we trained and evaluated three survival models each with and without polygenic risk information. Our results demonstrate a consistent improvement in predictive performance when polygenic risk scores are included, with the C-index improving from 0.690 to 0.709 in the best-performing model. The PRS emerged as the second strongest predictor of MM progression after M-spike concentration, underscoring the value of incorporating inherited genetic risk into clinical risk stratification. Across all model types, performance differences were modest, suggesting that the choice of model architecture is less critical than the quality and completeness of input features. While the use of synthetic data limits the generalizability of our findings, this work establishes a methodological foundation for systematic comparison of PRS-augmented survival models. Future work on real patient cohorts, longitudinal biomarker measurements, and externally validated polygenic scores will be essential to confirm these findings and translate them into clinical practice.

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